INTRODUCTION TO DATA MANAGEMENT

PROJECT REPORT

(Project Semester January-April 2025)

Railway Data Analysis

Submitted by

Harsh Raj

Registration No.:12306140

Programme and Section: B. Tech CSE & K23GS

Course Code: INT217

Under the Guidance of

Ms. Baljinder Kaur

Assistant Professor

School of Computer Science

UID:27952

Discipline of CSE/IT

Lovely School of Computer Science

Lovely Professional University, Phagwara

<u>CERTIFICATE</u>

This is to certify that Harsh Raj bearing Registration no. 12306140 has completed INT217 project titled, **“Railway Data Analysis”** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
School of ……………………………………………….
Lovely Professional University
Phagwara, Punjab.

Date:

DECLARATION

I, Harsh Raj, student of B. Tech under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 11-04-2025 Signature

Registration No. 12306140 Name of the student

Harsh Raj

Acknowledgment

I express my sincere thanks and deep gratitude to my mentor **Ms. Baljinder Kaur** , for his continuous guidance, valuable support, and encouragement throughout the completion of this project. I am also grateful to Lovely Professional University for providing the platform and resources to explore and enhance my skills in data analysis and visualization.

Table of Contents

1. Introduction

1.1 Project Overview

This project focuses on analysing UK train ride data using Microsoft Excel to build an interactive dashboard. The dashboard helps users explore key train journey metrics such as delays, refunds, revenue, ticket usage, and reasons for delay. Using Excel tools like PivotTables, slicers, and dynamic formulas, meaningful insights were derived to assist in transport efficiency and decision-making.

1.2 Importance of Train Data Analytics

Train transportation is vital in the UK, serving millions of commuters daily. Monitoring performance metrics such as delays, refund trends, and ticket class usage is essential for improving customer experience and operational reliability. This analysis empowers railway managers and passengers to understand travel trends and challenges.

---

2. Source of Dataset

2.1 Dataset Origin and Format

The dataset used was created to simulate UK train journey transactions. It includes transaction IDs, journey dates, stations, purchase details, train classes, delay times, and refund requests. The data was stored in .csv format and imported into Excel for cleaning and analysis.

2.2 Key Fields in the Dataset

Important columns in the dataset include:

- Transaction ID
- Date of Purchase
- Purchase Type
- Ticket Class
- Departure Station
- Arrival Destination
- Arrival Time & Actual Arrival Time
- Journey Status
- Reason for Delay
- Refund Request
- Price

---

3. Dataset Preprocessing

3.1 Added Helper Columns

- **Delay (Minutes)** = (Actual Arrival Time - Arrival Time) × 1440
- **Is Delayed** = IF Journey Status is "Delayed", then 1, else 0
- **Route** = Departure Station & " to " & Arrival Destination

3.2 Converting Data to Table Format

Data was converted into an Excel Table named Train Data by selecting the whole table and using the shortcut Ctrl + T, enabling the use of slicers and dynamic PivotTables for analysis.

---

4. Analysis Objectives & KPIs

4.1 Average Delay by Arrival Station

A PivotTable calculated the average delay per arrival station using the Delay (Minutes) column, helping identify which destinations experience the most frequent delays.

4.2 Total Refund Requests Over Time

A line chart and PivotTable displayed the number of refund requests over time, based on the Refund Request column.

4.3 Most Common Reason for Delay

A count of each Reason for Delay was visualized to see which operational issues occur most frequently.

4.4 Ticket Class Usage Breakdown

A pie chart showed the distribution of ticket class usage (Standard, First Class, etc.), helping to understand user preferences.

4.5 Top 5 Routes by Revenue

By summing Price for each unique route, the top 5 revenue-generating train routes were displayed using a bar chart.

4.6 Key Performance Indicators (KPIs)

The dashboard highlights the following KPIs:

- 💰 **Total Revenue**
- ⏱️ **Average Delay (Minutes)**

Each KPI updates dynamically based on slicer selections (e.g., station, ticket class).

---

5. Dashboard Design & Visualizations

5.1 Use of Pivot Tables and Slicers

PivotTables were used to aggregate metrics, and slicers were added to filter the dashboard by Departure Station, Ticket Class, Payment Method, and Date of Journey.

5.2 Dashboard Layout

The layout includes two KPI cards at the bottom, slicers to the left and middle bottom, and charts arranged in a clean grid in the centre of the sheet.

5.3 Description of Charts

Charts used:

- Bar Chart: Average Delay per Arrival Station
- Line Chart: Refund Requests Over Time
- Pie Chart: Ticket Class Breakdown
- Column Chart: Most Common Reasons for Delay
- Bar Chart: Top 5 Routes by Revenue

---

6. Analysis Results

6.1 Delay and Punctuality Insights

Trains to major destinations like [Example Station] show the highest delays. The average system-wide delay is approximately [X] minutes.

6.2 Refund Trends and Passenger Behaviour

Refund requests are highest during peak travel periods. A higher % of refund requests correlates with delays of over 10 minutes.

6.3 Revenue Generation by Routes

Routes from [Station A to Station B] generate the most revenue, indicating high demand or higher fares.

6.4 Ticket Class Preferences

Most passengers travel in Standard class, but First Class accounts for higher revenue per ride.

6.5 Visual Trends and Observations

Data visualizations revealed patterns of delays and refund spikes that align with specific times of day and payment types.

---

7. Conclusion

This project successfully demonstrates the power of Excel for building data-driven dashboards. By using PivotTables, slicers, and calculated columns, we analyzed UK train ride data to understand travel efficiency, passenger experience, and revenue generation.

---

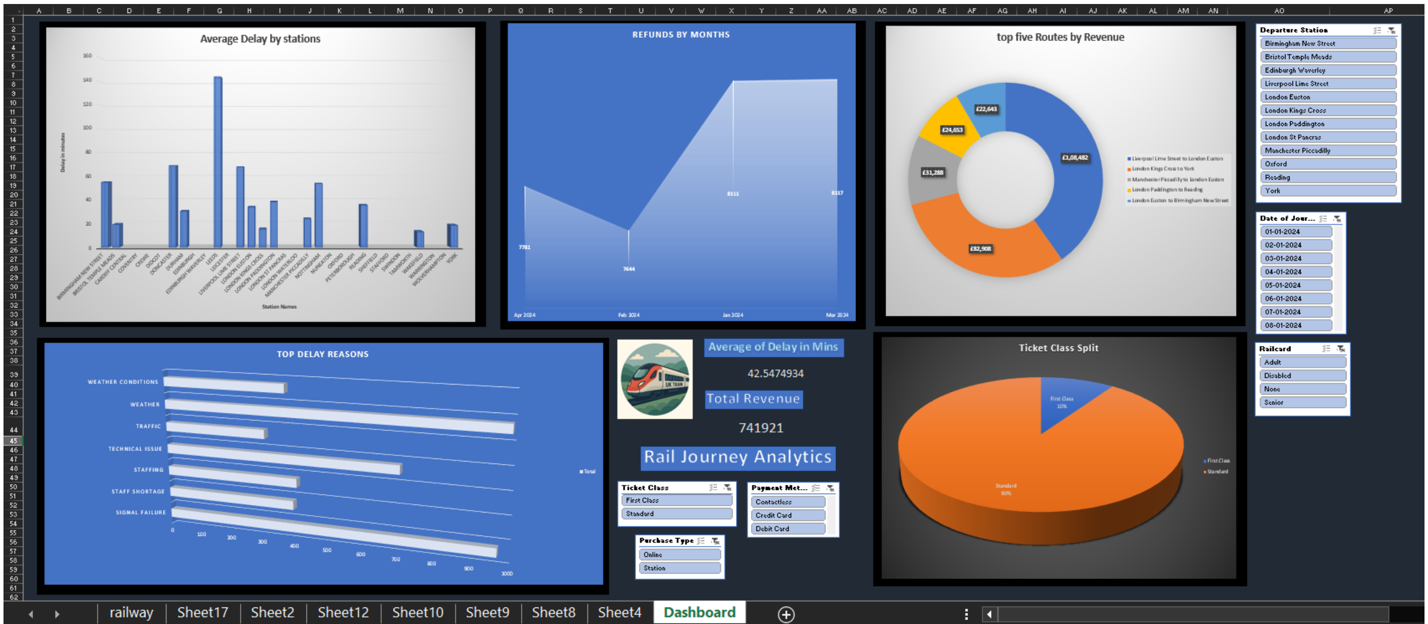
8. Future Scope

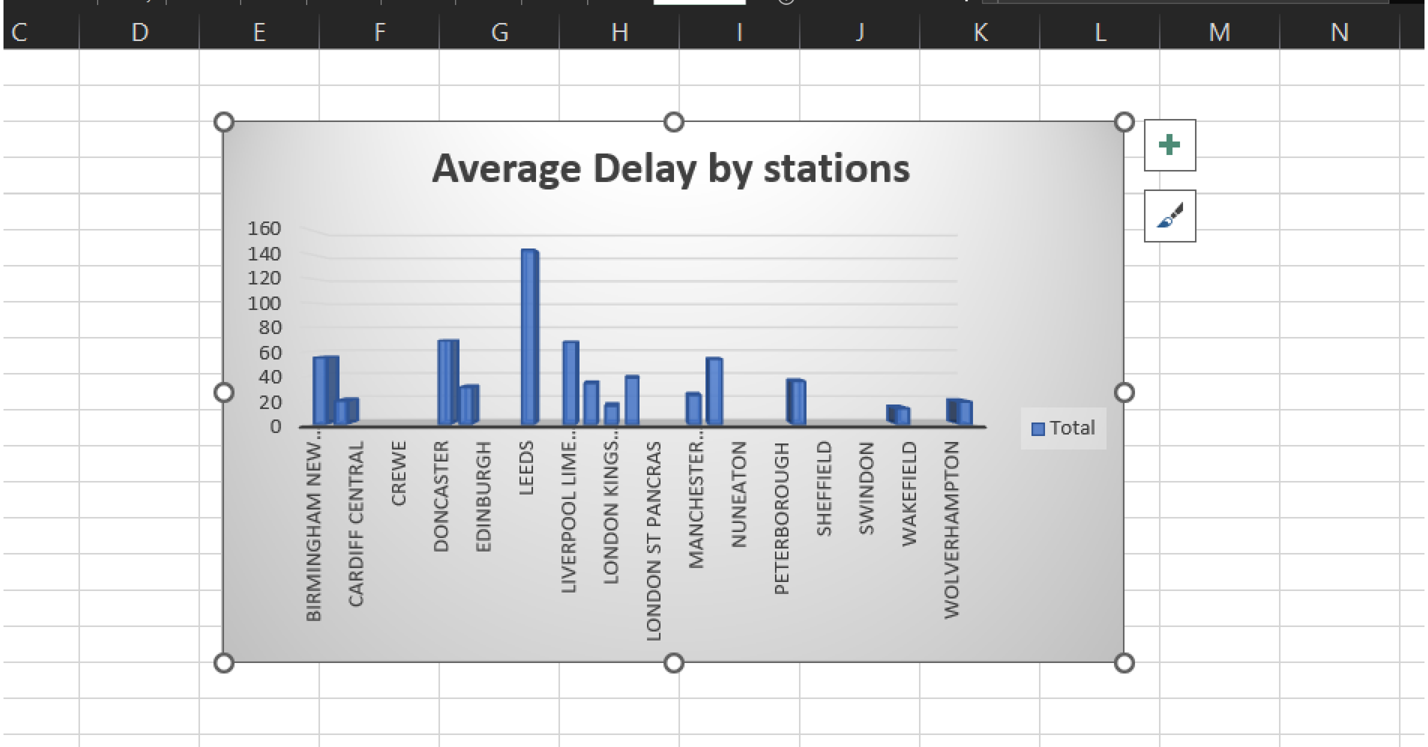
- Use Power BI for more advanced interactive dashboards
- Add maps to visualize station-wise delays
- Include forecasting models for refund rates and revenue
- Automate dashboard updates with real-time data feeds

---

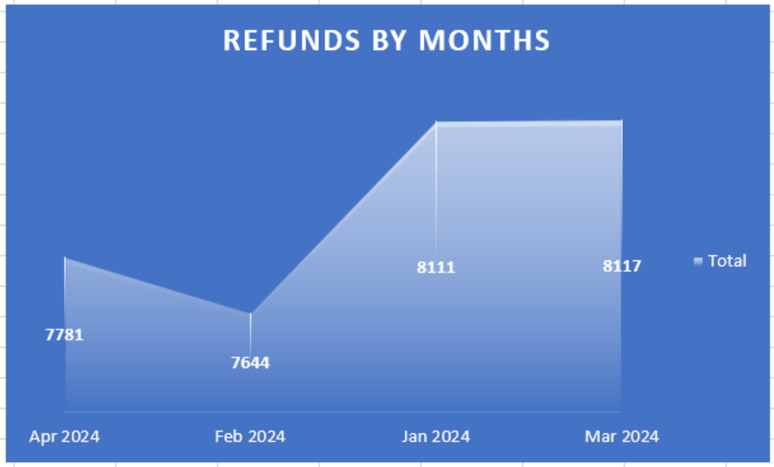
9. References

- Microsoft. PivotTables in Excel. https://support.microsoft.com
- UK Open Transport Data (fictional, or your actual source)
- Excel Help: https://exceljet.net

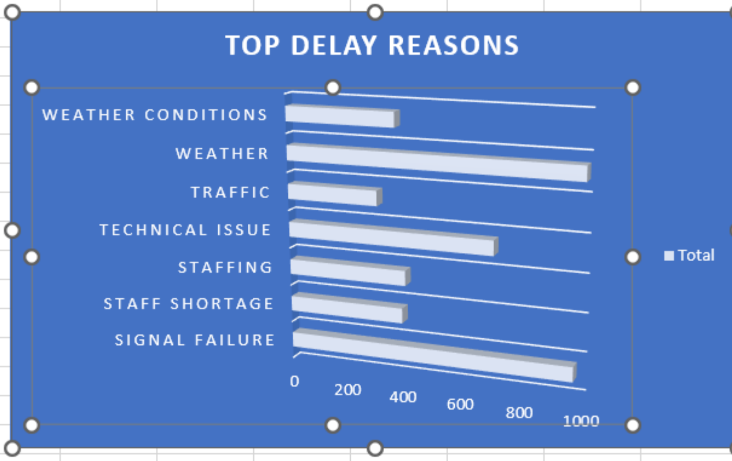

Average Delay by stations
REFUNDS BY MONTHS
top five Routes by Revenue
TOP DELAY REASONS
Average of Delay in Mins
42.5474934
Total Revenue
741921
Rail Journey Analytics
Ticket Class Split
railway
Sheet17
Sheet2
Sheet12
Sheet10
Sheet9
Sheet8
Sheet4
Dashboard


Average Delay by stations
160
140
120
100
80
60
40
20
0
Total
BIRMINGHAM NEW...
CARDIFF CENTRAL
CREWE
DONCASTER
EDINBURGH
LEEDS
LIVERPOOL LIME...
LONDON KINGS...
LONDON ST PANCRAS
MANCHESTER...
NUNEATON
PETERBOROUGH
SHEFFIELD
SWINDON
WAKEFIELD
WOLVERHAMPTON

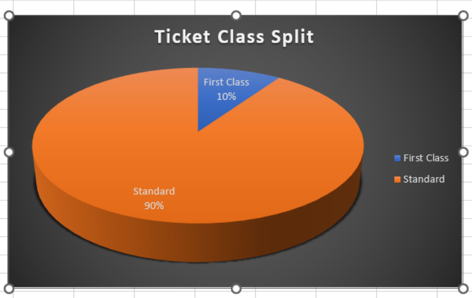
Row Labels	Count of Refund Request
Apr 2024	7781
Feb 2024	7644
Jan 2024	8111
Mar 2024	8117
Grand Total	**31653**



Row Labels	Count of Transaction ID
Signal Failure	970
Staff Shortage	399
Staffing	410
Technical Issue	707
Traffic	314
Weather	995
Weather Conditions	377
Grand Total	**4172**



Row Labels	Count of Transaction ID
First Class	3058
Standard	28595
Grand Total	**31653**



Payment Method

- Contactless
- Credit Card
- Debit Card

Railcard

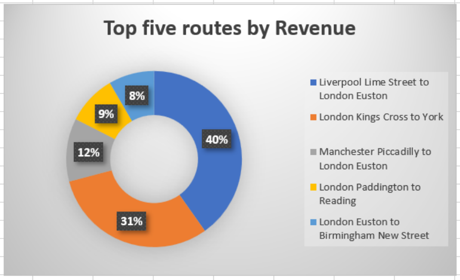
- Adult
- Disabled
- None
- Senior

Date of Journey

- 01-01-2024
- 02-01-2024
- 03-01-2024
- 04-01-2024
- 05-01-2024
- 06-01-2024
- 07-01-2024
- 08-01-2024

A3 Row Labels

	Row Labels	Sum of Price
3	**Row Labels**	**Sum of Price**
4	Liverpool Lime Street to London Euston	108482
5	London Kings Cross to York	82908
6	Manchester Piccadilly to London Euston	31288
7	London Paddington to Reading	24653
8	London Euston to Birmingham New Street	22643
9	**Grand Total**	**269974**



Ticket Class
First Class
Standard

Railcard
Adult
Disabled
None
Senior

Purchase Type
Online
Station

T10 =TEXT([@[Date of Journey]], "mmm yyyy")

	Transaction ID	Date of Purchase	Time of Purchase	Purchase Type	Payment Method	Railcard	Ticket Class	Ticket Type	Price	Departure Station	Arrival Destination	Date of Journey	Departure Time
2	da8a6ba8-b3dc-4677-b176	08-12-2023	12:41:11	Online	Contactless	Adult	Standard	Advance	43	London Paddington	Liverpool Lime Street	01-01-2024	11:00:00
3	b0cdd1b0-f214-4197-be53	16-12-2023	11:23:01	Station	Credit Card	Adult	Standard	Advance	23	London Kings Cross	York	01-01-2024	09:45:00
4	f3ba7a96-f713-40d9-9629	19-12-2023	19:51:27	Online	Credit Card	None	Standard	Advance	3	Liverpool Lime Street	Manchester Piccadilly	02-01-2024	18:15:00
5	b2471f11-4fe7-4c87-8ab4	20-12-2023	23:00:36	Station	Credit Card	None	Standard	Advance	13	London Paddington	Reading	01-01-2024	21:30:00
6	2be00b45-0762-485e-a7a3	27-12-2023	18:22:56	Online	Contactless	None	Standard	Advance	76	Liverpool Lime Street	London Euston	01-01-2024	16:45:00
7	4e1dcd88-3d95-44ef-99fa	30-12-2023	07:56:06	Online	Credit Card	None	Standard	Advance	35	London Kings Cross	York	01-01-2024	06:15:00
8	1c74479d-85a4-4ba1-a607	31-12-2023	00:02:01	Station	Credit Card	Adult	Standard	Advance	2	London Euston	Oxford	01-01-2024	22:30:00
9	febf8dab-f808-46fa-bf2b	31-12-2023	01:35:18	Station	Contactless	Disabled	Standard	Advance	2	Liverpool Lime Street	Manchester Piccadilly	01-01-2024	00:00:00
10	01df916f-4291-41ec-a37d	31-12-2023	01:43:09	Station	Credit Card	None	Standard	Advance	37	London Euston	York	01-01-2024	00:00:00
11	a8cedba7-1923-459d-b046	31-12-2023	03:05:52	Online	Credit Card	None	Standard	Advance	13	London Paddington	Reading	01-01-2024	01:30:00
12	b3e5ca7d-e76c-49f2-b49f	31-12-2023	03:26:37	Online	Contactless	None	Standard	Advance	8	York	Durham	01-01-2024	01:45:00
13	6c63f7ac-d590-4356-9eaa	31-12-2023	03:52:11	Online	Contactless	Adult	Standard	Advance	8	London Paddington	Reading	01-01-2024	02:15:00
14	2e7add75-566a-41aa-9468	31-12-2023	05:55:22	Online	Contactless	None	Standard	Advance	3	Manchester Piccadilly	Liverpool Lime Street	01-01-2024	04:15:00
15	7ed9b545-eb6f-49b2-9b5a	31-12-2023	06:44:35	Online	Contactless	None	Standard	Advance	3	Manchester Piccadilly	Liverpool Lime Street	01-01-2024	05:00:00
16	2e05e2a6-88a8-40fb-bacc	31-12-2023	08:05:50	Online	Credit Card	Disabled	Standard	Advance	15	Birmingham New Street	London St Pancras	01-01-2024	06:30:00
17	8a18d3b4-995e-49bf-93a3	31-12-2023	08:16:53	Online	Credit Card	None	Standard	Advance	13	London Paddington	Reading	01-01-2024	07:45:00
18	7493a611-342a-4b17-90dc	31-12-2023	08:23:15	Online	Credit Card	None	Standard	Advance	13	London Paddington	Reading	01-01-2024	07:45:00
19	054676ac-a976-4909-a26e	31-12-2023	09:09:20	Online	Credit Card	None	Standard	Advance	8	London St Pancras	Birmingham New Street	01-01-2024	07:30:00
20	6b62b452-c491-468d-b39c	31-12-2023	09:12:21	Online	Credit Card	None	Standard	Advance	35	London Kings Cross	York	01-01-2024	07:30:00
21	85e38992-6c6c-4569-914e	31-12-2023	10:42:22	Online	Credit Card	None	Standard	Advance	35	London Kings Cross	York	01-01-2024	09:00:00
22	8dfbf0fc-aea0-424f-b30e	31-12-2023	11:57:15	Station	Debit Card	Adult	Standard	Advance	7	Birmingham New Street	Manchester Piccadilly	01-01-2024	11:15:00
23	a478f358-044d-4000-b70e	31-12-2023	12:11:47	Station	Credit Card	Disabled	Standard	Advance	2	Liverpool Lime Street	Manchester Piccadilly	01-01-2024	10:30:00
24	d4ea6940-7228-41ef-9eae	31-12-2023	13:33:39	Online	Credit Card	Senior	Standard	Advance	23	London Kings Cross	York	01-01-2024	09:30:00
25	89f2160c-666b-4925-86b6	31-12-2023	14:23:09	Station	Contactless	None	Standard	Advance	7	London Euston	Birmingham New Street	01-01-2024	12:45:00
26	fd00a134-7056-441c-919f	31-12-2023	14:53:44	Online	Credit Card	None	Standard	Advance	7	London Euston	Birmingham New Street	01-01-2024	13:15:00
27	842da93c-b820-42dc-ad4f	31-12-2023	15:19:53	Online	Contactless	None	Standard	Advance	86	Manchester Piccadilly	London Paddington	01-01-2024	13:45:00
28	74462231-5241-46f4-8328	31-12-2023	15:53:46	Online	Credit Card	Senior	First Class	Advance	34	Oxford	Bristol Temple Meads	01-01-2024	14:15:00
29	7267c284-5f19-41ef-8350	31-12-2023	16:44:05	Station	Credit Card	None	Standard	Advance	13	London Paddington	Reading	01-01-2024	15:00:00
30	3c375a4c-7ba3-45e9-a9cd	31-12-2023	18:01:58	Online	Contactless	None	Standard	Advance	7	London Euston	Birmingham New Street	01-01-2024	16:30:00

railway | Sheet2 | Sheet12 | Sheet10 | Sheet9 | Sheet8 | Sheet4 | Dashboard